Chinese Reproduction: Added Qwen3-4B Runs

Same FUR setting

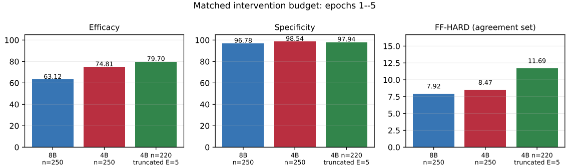
Dataset	Chinese C-Eval subset
Unlearning	NPO-KL, stepwise
LR / seed	10^{-5} / 1001
Specificity split	20 held-out samples
Updated params	FF2 only
CoT style	Visible Chinese CoT, no thinking mode

Runs compared

Run	CoTs	Targets	Epochs
Qwen3-8B	250	230	5
Qwen3-4B	250	230	5
Qwen3-4B	220	200	**10**

Important. The $n = 220$ Qwen3-4B run is not a direct duplicate: it uses **10 unlearning epochs**. We treat it as an intervention-strength extension, not as a pure model-size comparison.

Three-Run Results: Matched Five-Epoch View



Paper-style metrics through epoch 5

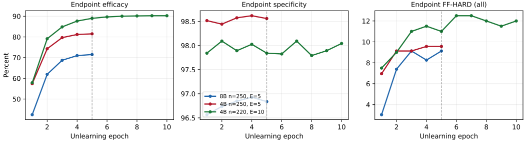
Run	Eff	Spec	FF-HARD
8B, $n = 250$	63.12	96.78	7.92%
4B, $n = 250$	**74.81**	**98.54**	8.47%
4B, $n = 220$, E5 cut	**79.70**	97.94	**11.69%**

Under the matched $n = 250, E = 5$ setting, Qwen3-4B has much higher efficacy than Qwen3-8B, but only a small FF-HARD gain.

On the common eligible subset (158 questions), FF-HARD is 7.59% for 4B vs. 6.96% for 8B. The strongest signal is **intervention strength**, not a large faithfulness gap.

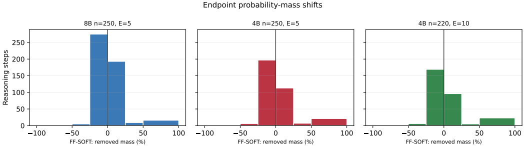
Extended 4B Run: Epoch Effect and Probability Shifts

Unlearning trajectories



E5 $\rightarrow$ E10 for 4B $n = 220$: Eff 88.99 $\rightarrow$ 90.25, FF-HARD(all) 11.00% $\rightarrow$ 12.00%.

Endpoint FF-SOFT distribution



Ten epochs move more probability mass away from the original answer, but the gain saturates after roughly epoch 6.

Takeaway. **Qwen3-4B $n = 220$, $E = 10$ is an epoch-strength extension.** It supports stronger intervention effects, but should not be merged into the standard $E = 5$ model ranking.